

Facet Control for Trap-State Suppression in Colloidal Quantum Dot Solids

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Trap states in colloidal quantum dot (QD) solids significantly affect the performance of QD solar cells, because they limit the open-circuit voltage and short circuit current. The {100} facets of PbS QDs are important origins of trap states due to their weak or missing passivation. However, previous investigations focused on synthesis, ligand exchange, or passivation approaches and ignored the control of {100} facets for a given dot size. Herein, trap states are suppressed from the source via facet control of PbS QDs. The {100} facets of ≈ 3 nm PbS QDs are minimized by tuning the balance between the growth kinetics and thermodynamics in the synthesis. The PbS QDs synthesized at a relatively low temperature with a high oversaturation follow a kinetics-dominated growth, producing nearly octahedral nanoparticles terminated mostly by {111} facets. In contrast, the PbS QDs synthesized at a relatively high temperature follow a thermodynamics-dominated growth. Thus, a spherical shape is preferred, producing truncated octahedral nanoparticles with more {100} facets. Compared to PbS QDs from thermodynamics-dominated growth, the PbS QDs with less {100} facets show fewer trap states in the QD solids, leading to a better photovoltaic device performance with a power conversion efficiency of 11.5%.

1. Introduction

Colloidal quantum dots (QDs) are promising semiconductor materials, due to their size-tunable bandgap, solution processability, and adjustable surface chemistry,^[1] for applications in devices including light-emitting diodes,^[2–3] photodetectors,^[4] lasers,^[5] transistors,^[6] and solar cells.^[7] With the significant progress in synthesis, ligand exchange, and device structure, the performance of PbS QD solar cells has been improved rapidly, achieving a certified power conversion efficiency (PCE) of 12.3%.^[8–14] In addition, PbS QD solar cells also show excellent stability, demonstrated by nonencapsulated devices exhibiting no loss in PCE for over 150 days of storage in air.^[11] Therefore, PbS QDs are a very promising potential photovoltaic material for low-cost and high-efficiency solar cells.

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DOI: 10.1002/adfm.202000594

The steady growth of the PCE of QD solar cells is mainly attributed to the passivation of trap states in QD solids, as the trap states substantially limit the device performance.^[15–18] Deep trap states assist the recombination of photocarriers, reducing both the open-circuit voltage (V_{OC}) and short circuit current density (J_{SC}). Shallow trap states slow down the transport of carriers, limiting the efficient extraction of photocarriers. Additionally, shallow trap states increase the bandtail states, reducing the V_{OC} and might causing band misalignment.^[19] Therefore, different strategies were developed in order to suppress trap states of QDs. Tang et al. introduced atomic ligand, halide anions, to passivate surface defects in QD solids.^[8] Ip et al. developed a solution-based approach to introduce halide anions, giving rise to organic–inorganic hybrid passivation and further improvement of the device performance.^[20] Liu et al.^[19] and Aqoma et al.^[10] developed a solution-phase ligand exchange method, leading to efficient ligand exchange and improved passivation of PbS QD solids. In our previous work, we developed in situ halide passivation via diffusion-controlled growth^[9] and cation (Cd^{2+})–anion (Cl^-) hybrid passivation via cation exchange synthesis,^[21] resulting in effective surface passivation and improved performance of QD solar cells.

Unbound surface sites before and after the ligand exchange are considered to generate trap states, which are attacked by other species, such as oxygen or water. The binding of ligands strongly depends on crystallographic facets, which are well known to be {111} and {100} facets for lead chalcogenide QDs.^[22–23] The oleate ligands predominately bind to the {111} facets, which are terminated with Pb atoms. The {100} facets have a checkerboard arrangement of alternating Pb^{2+} cations and S^{2-} anions and they are usually not capped with organic ligands due to the weak binding energy of ligands to these facets.^[24–26] Wang and co-workers^[22] showed that oleic acid (OA) can be adsorbed on the {100} facets, however, the bond is very weak and the ligands can be easily removed (such as gentle purification) from these facets. Consequently, the {100} facets are prone to oxidation in air. The facet-specific ligand adsorption explains the size-dependent air-stability. As the area of {100} facets increases with dot size, the stability decreases.^[25] The preferential ligand passivation on the {111} facets also accounts for the fact that PbSe QDs have poorer air-stability than PbS QDs. The {111} facets are terminated with Pb atoms both in PbS and PbSe QDs, while the difference is the chalcogenide anions on the {100} facets. A Se atom has a smaller electronegativity than a S atom. Consequently, PbSe QDs are more prone to oxidation than PbS QDs.

The {100} facets are important origins of trap states due to their weak or missing passivation. First, the {100} facets might introduce trap states before ligand exchange via oxidation or reaction with other species. Second, the bare {100} facets might initiate heterogeneous fusion during ligand exchange, leading to nonuniform aggregation, which is detrimental to QD solar cells. For example, it increases the energetic inhomogeneity, leading to a poor carrier transport. Moreover, it increases the loss of V_{OC} through broadening of the bandtail state distribution and it might lead to a band misalignment. Third, the {100} facets might be insufficiently passivated after typical ligand exchange. For this reason, Kim et al.^[26] added sodium acetate in the common ligand exchange, rendering Na^+ -passivation to the

{100} facets. Therefore, it is desirable to reduce the {100} facets for a given size of PbS QDs, which is ignored in previous investigations. In the former synthesis routes, efforts were focused on narrowing size distributions and improving surface passivation, without facet control. The ligand exchange and passivation strategies mentioned above were ways to remedy the unbound surface sites with given terminated facets. Thus, it would be preferable to suppress trap states from the source instead of repairing these defects afterward.

In the present work, we have managed to control the facets by tuning the balance between kinetic and thermodynamic growth conditions via a fine control of the growth conditions. For ≈ 3 nm PbS QDs, nearly octahedral nanoparticles are obtained from a kinetics-dominated growth at a relatively low growth temperature with a high oversaturation. On the other hand, truncated octahedral nanoparticles are produced from a thermodynamics-dominated growth at a relatively high temperature with a low oversaturation. The nearly octahedral nanoparticles are terminated with almost only {111} facets, while the truncated octahedral ones have more {100} facets. Consequently, compared to the thermodynamically grown PbS QDs, the kinetically grown ones exhibit fewer trap states in QD solids and consequently, a better device performance is achieved with a PCE of 11.5%.

2. Results and Discussion

2.1. Facet Control of PbS QDs

The formation of QDs can be separated into two steps: nucleation and the subsequent growth from the seeds.^[27] The growth process affects the final geometry and terminates the facets, which is mainly controlled by two factors: the surface energies of different crystallographic facets (including the effect of ligand adsorption) and the growth regime (kinetics or thermodynamics). When a sufficient supply of thermal energy and a low flux of monomer are provided, thermodynamic growth is possible, while kinetic growth occurs at low temperatures with a high flux of monomer.^[28] Under thermodynamic growth conditions, seeds grow isotropically and spheres are favorable as they are the thermodynamically most stable shape with the lowest overall surface energy. In contrast, under kinetic growth conditions, seeds grow anisotropically with preferential facets that have higher surface energies. Therefore, it is possible to control the terminating facets by changing the growth parameters, such as the growth temperature and the monomer concentration. For example, Peng et al.^[29–30] controlled the growth regime of CdSe nanocrystals (NCs) by changing the monomer concentration. At a fixed growth temperature, thermodynamic growth was favorable under a low monomer concentration, thus the CdSe NCs grew isotropically resulting in a spherical shape.^[30] Under a high monomer concentration, the CdSe NCs followed a kinetic growth and grew anisotropically along the *c*-axis of the wurtzite structure, producing rod-shaped CdSe NCs.^[30] Additionally, Cheon and co-workers^[31] tuned the growth regime of large PbS NCs (tens of nanometers) by changing the growth temperature. At a high temperature (≈ 300 °C), thermal energy was sufficient to support the thermodynamic growth, thus

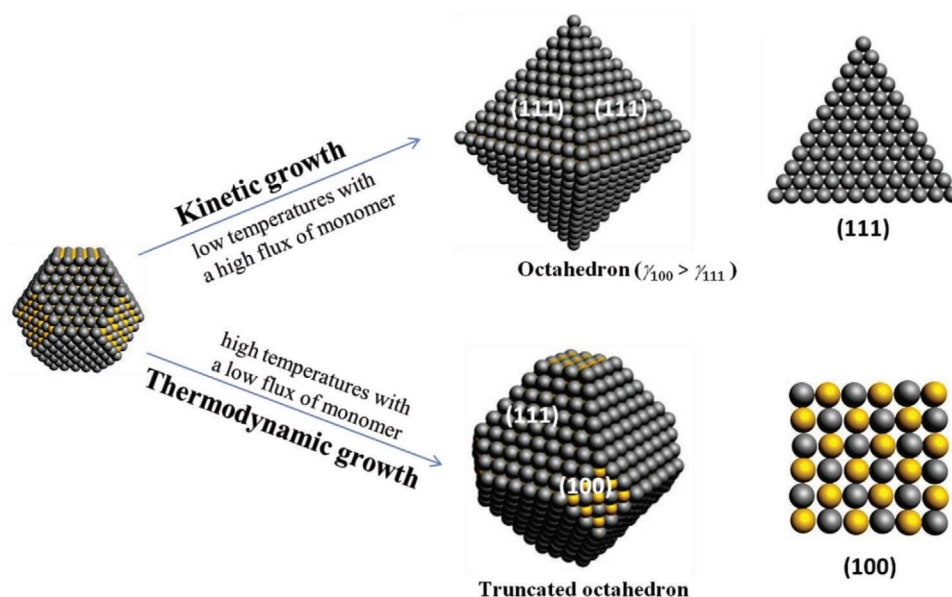


Figure 1. Controlling the terminating facets of PbS QDs via a balance of the growth regimes between kinetic (top) and thermodynamic growth conditions (bottom). The black and yellow spheres represent Pb and S atoms, respectively.

isotropic growth of the PbS NCs was observed.^[31] At a low temperature ($\approx 140^\circ\text{C}$), the growth followed the kinetic regime and multipod structures or star-shaped PbS NCs were produced via anisotropic growth.^[31]

Herein, we choose the growth regime as the tool to control the terminating facets of ≈ 3 nm PbS QDs without changing the ligands and the reaction system, as shown in **Figure 1**. In brief, cation exchange is adopted as the synthetic method based on our previous work, using ZnS QDs as the starting material. First, ZnS QDs are converted to PbS QDs serving as the seeds for the following growth. Then, the subsequent growth is controlled by adjusting the temperature and monomer concentration. Since the size of the seeds is strongly related to the size of starting ZnS QDs, extremely small PbS seeds are achievable by using extremely small ZnS QDs, allowing for a large room for the growth to obtain a size of ≈ 3 nm with an exciton peak of ≈ 900 nm. Because the target of the present study is small PbS QDs (≈ 3 nm), the growth temperature should not be very high. Therefore, it is impossible to get pure thermodynamic growth, however, tuning the balance between kinetic and thermodynamic growths is possible. For the kinetics-dominated growth regime, the synthesis is performed at a relatively low temperature (60°C) with a high monomer concentration which is realized by the dropwise addition of extremely small ZnS QDs. For the thermodynamics-dominated growth regime, a relatively high temperature (130°C) is applied without further addition of extremely small ZnS QDs. The synthesis of the two growth regimes are schematically shown in Figure S1 in the Supporting Information.

Under kinetic growth conditions, the growth rate is exponential in the surface energy;^[27,28] thus, the termination of nanocrystal facets with smaller surface energies is strongly favored. It is well known that PbS NCs generally nucleate as tetradecahedron seeds as shown in Figure 1, exposing six {100} facets and eight {111} facets.^[28,31–32] Therefore, the difference in

surface energy between the {100} and {111} facets determines the terminating facets. We calculated the surface energies of {100} and {111} facets with ligands using density functional theory (DFT). The organic ligand is oleylamine (OLA) during the growth of PbS QDs, although oleic acid replaces the OLA when the growth is finished. Because PbCl_2 is used as the lead source, Cl^- also is adsorbed on the surface as the inorganic ligand. Previous studies have shown that alkylamine ligands preferentially bind to the {111} facets which are Pb-rich,^[33] while Cl^- binds to the {100} facets forming atomically thin adlayers.^[34] In addition to ligands, steric hindrance also dictates the surface energy. Jeong and co-workers have shown that the steric hindrance is mitigated for ultrasmall PbS QDs (≤ 4 nm), reducing the surface energy.^[25] Considering the facet-specific ligands and the steric hindrance, the surface energies (γ) of {100}, ultrasmall {111} (corresponding to the case here), and large {111} facets are calculated to be 4.39, -7.49 , and 38.8 eV nm^{-2} , respectively, as shown in **Figure 2** (the details can be found in the Supporting Information).

Because $\gamma_{100} > \gamma_{111}$, ≈ 3 nm PbS QDs will terminate only with {111} facets under kinetics-dominated growth. Thus the shape of the QDs is expected to an octahedron. Considering the small size and the reconstruction of atoms on the surface, a perfect octahedron would not be obtained in reality, however, a nearly octahedron shape of the QDs is more likely. On the other hand, under thermodynamics-dominated growth, the ≈ 3 nm PbS QDs will adopt a shape of truncated octahedron, which is closer a sphere, with terminating facets being {111} and {100}. Transmission electron microscope (TEM) is a direct characterization to show the difference in shape. Unfortunately, ≈ 3 nm PbS QDs are too small to identify the difference between octahedrons and truncated octahedrons in TEM images, as shown in Figure S2 in the Supporting Information. However, for large PbS QDs, the effect of the growth regime on the shape (terminating facets) is easily observed, as

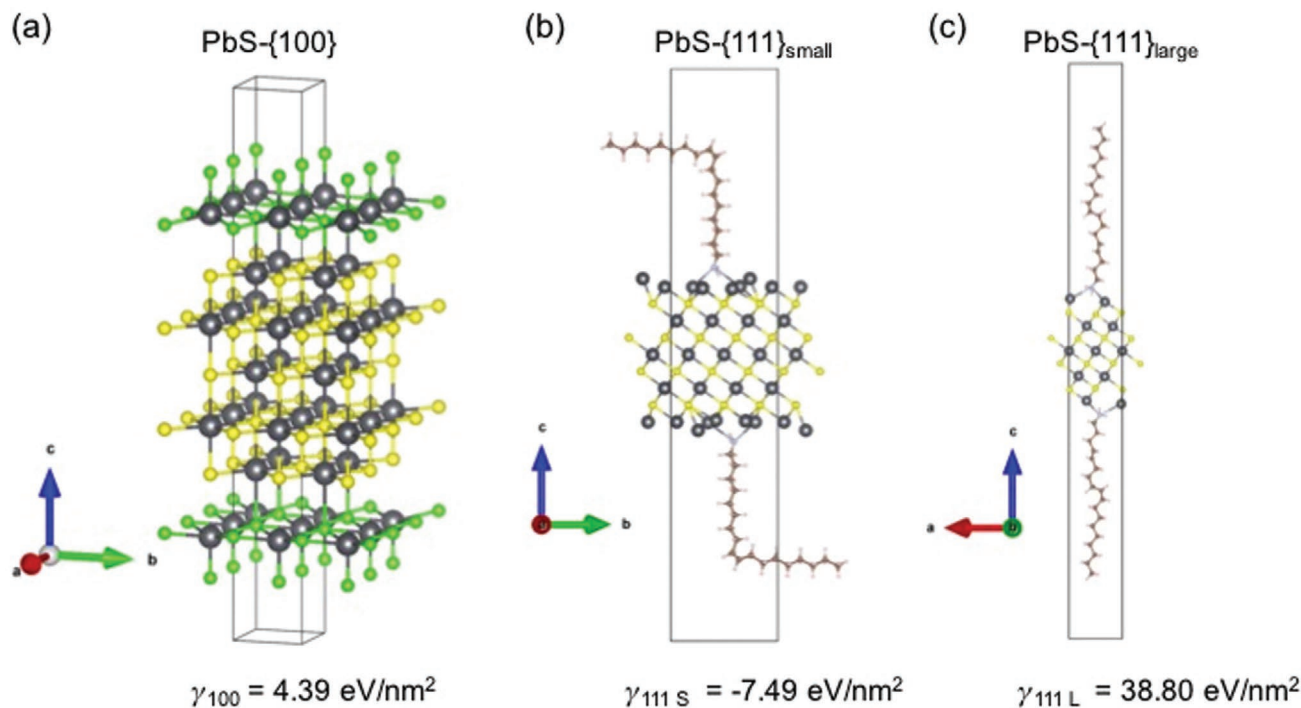


Figure 2. Calculation of surface energies of PbS QDs with ligands. a) {100} facets with a layer of PbCl₂. b) OLA passivated ultrasmall {111} facets with steric hindrance mitigated. c) OLA passivated large {111} facets with steric hindrance. The {111} facets are Pb-rich and one OLA molecular coordinates to two Pb atoms due to the lone pair of electrons on N.

shown in Figure S3 in the Supporting Information. PbS QDs ($\approx 10 \text{ nm}$) obtained under kinetics-dominated growth conditions (low temperature and high monomer flux) have a shape that resembles cubes (Figure S3a, Supporting Information), while those ($\approx 10 \text{ nm}$) obtained under thermodynamics-dominated growth conditions (high temperature and low monomer flux) adopt a shape that resembles spheres (Figure S3b, Supporting Information).

The atoms on the {111} and {100} facets of PbS QDs are different. The {111} facets are terminated with Pb atoms, while the {100} facets are composed of Pb and S atoms with equal ratios. Therefore, for a given size, the octahedral PbS QDs should have a higher Pb/S ratio than the truncated octahedral ones. This difference is confirmed by the element analysis measured by X-ray photoelectron spectroscopy (XPS). The PbS QDs grown under the kinetics-dominated regime exhibit a Pb/S ratio of 1.33, while those obtained from thermodynamics-dominated growth show a ratio of 1.10 (Figure S4 and Table S1, Supporting Information).

In order to identify the difference in shape and thereby the terminating facets, we perform grazing-incidence small-angle X-ray scattering (GISAXS) experiments on as-deposited QD films without ligand exchange treatment, because the shape of nanoparticles is the key factor for the stacking behavior in the QD assemblies. For the truncated octahedral shape, which is close to a sphere, the QDs will naturally adopt a face-centered cubic (FCC) stacking behavior, driven by the system entropy.^[35] For the octahedral shape, the QDs are prone to a stack with a body-centered cubic/tetragonal (BCC/BCT) layout according to the Wigner–Seitz close packed construction, which is driven

by the decrease of the total surface energy of the particle system.^[35–36] The GISAXS results are in good agreement with the expected stacking behavior for the different QD shapes (see Figure 3). QDs synthesized under thermodynamics-dominated growth conditions are denoted as truncated-octahedral (*to*)-QDs according to their expected shape and those synthesized under kinetics-dominated growth conditions are denoted as nearly octahedral (*o*)-QDs. 2D GISAXS data of spin-coated QD assemblies are shown in Figure 3a,b, in which the calculated corresponding Bragg peaks from the superlattice models are shown as white dots. The *to*-QD superlattice reveals a FCC particle stacking behavior, which is due to the shape of a *to*-QD being similar to a sphere, especially with the ligands capping on the surface of the QDs. The superlattice has a (111)_{SL} plane parallel to the substrate, and the superlattice parameters from the calculation are determined as $a = b = c = 8.8 \text{ nm}$, and $\alpha = \beta = \gamma = 90^\circ$ (Figure 3c). The slight shift of (200)_{SL} peak position from the predicted position can be explained by a deviation from a close packing of hard spheres by the presence of the ligand shell which introduces a limited softness of the QDs. In contrast, the *o*-QDs demonstrate a distinct different stacking behavior compared with *to*-QDs, as seen by the different scattering pattern (Figure 3b). A BCT superlattice model is found to fit the scattering pattern best by comparing the scattering peak positions in the real 2D data and in calculations. The superlattice parameters are $a = b = 9.0 \text{ nm}$, $c = 10.6 \text{ nm}$, and $\alpha = \beta = \gamma = 90^\circ$ as seen in Figure 3d. The (110)_{SL} plane is supposed to be parallel to the substrate. These findings also match well with previous findings about the stacking behavior of nano-octahedrons.^[37–38] Notably, even although *o*-QDs have

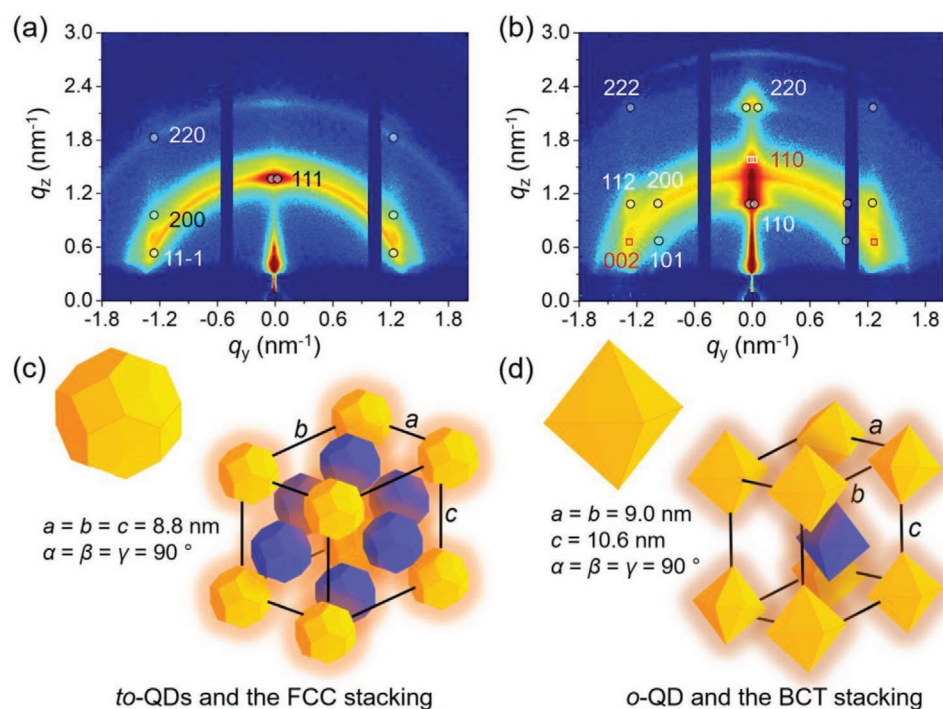


Figure 3. Effect of particle shape on the packing behavior in ≈ 3 nm PbS QD assemblies. 2D GISAXS data of the superlattice formed by a) truncated octahedral QDs (*to*-QDs) and b) octahedral QDs (*o*-QDs). The white dots in the GISAXS data are calculated scattering peaks based on standard superlattice models. Diagrams for c) FCC and d) BCT superlattices with different lattice parameters.

larger {111} facet areas that allow binding of more ligands than the *to*-QDs, the shape of the QDs still plays an important role in the achieved close packed configuration of the QDs, since ligands of the QDs will collapse in fully evaporated solvent conditions.^[39] Notably, this effect also explains the superlattice distortion in the *to*-QD-based FCC superlattice, being a synergy effect with the shape of the QDs.

2.2. Trap-State Suppression in QD Solids

As evidenced by the analyses of the effect of the growth regime on the final shape for large PbS QDs, the Pb/S ratio and the effect of QD (≈ 3 nm) shape on the packing manner, the kinetics-dominated growth produces PbS QDs with a nearly octahedral shape while the thermodynamics-dominated growth results in truncated octahedral PbS QDs. The nearly octahedral PbS QDs have smaller {100} facets than the truncated octahedral ones. Thus octahedral PbS QDs are expected to have fewer trap states, which is verified by optical characterizations. As shown in **Figure 4a**, the as-synthesized *o*-PbS QDs show a higher photoluminescence quantum yield (PL QY) of 50% than the *to*-QDs (37%), indicating that the *o*-PbS QDs have a better surface passivation and fewer trap states. The smaller {100} facets could avoid fusion, etching, and oxidation during the liquid ligand exchange.^[26] As shown in **Figure 4b**, the *to*-PbS QD solids show a broader PL spectrum than the *o*-QD solid and have stronger emission on the low energy side, suggesting severer aggregation for the *to*-PbS QDs. Additionally, the *o*-PbS QD solids have a longer PL lifetime than the *to*-QD solids as shown in **Figure 4c**,

confirming that the smaller {100} facets contribute to better protection of QDs during and after the ligand exchange. In order to explore energetic disorder in the PbS QD solids, we adopt the tail state absorption to estimate the Urbach energy.^[26,40] The *o*-PbS QD solid exhibits a lower Urbach energy of 28.6 meV than the *to*-QD solid (38.5 meV), indicating a lower trap density and fewer tail states in the QD solids with smaller {100} facets. Furthermore, the *o*-PbS QD solids show a shallower bandtail, demonstrated by the smaller redshift of the transient bleach peak in the spectrotemporal transient absorption (TA) map as shown in **Figure 4e,f**. This is in line with the PL and Urbach energy characterizations and a smaller V_{OC} deficit is expected for the *o*-PbS QD-based solar cells. Additionally, XPS analysis shows that the *o*-PbS QD solids have less oxygen contamination than the *to*-QD solids (**Figure S5**, Supporting Information), which is another demonstration of the suppressed {100} facets and trap states for the *o*-PbS QDs.

2.3. QD Solar Cells

Solar cells are fabricated based on the two types of PbS QDs using a device structure as shown in **Figure 5a**. The light-absorbing layers have a thickness of ≈ 350 nm (**Figure 5b**). Thanks to the fewer trap states and less energetic disorder in *o*-PbS QD solids, solar cells based on these QD solids show better performance than those based on *to*-PbS QD solids, as shown in **Figure 5c**. The *o*-PbS QD solar cells exhibit a highest efficiency of 11.5% with a short-circuit current density (J_{SC}) of 27.1 mA cm^{-2} , a V_{OC} of 0.64 V, and a fill factor

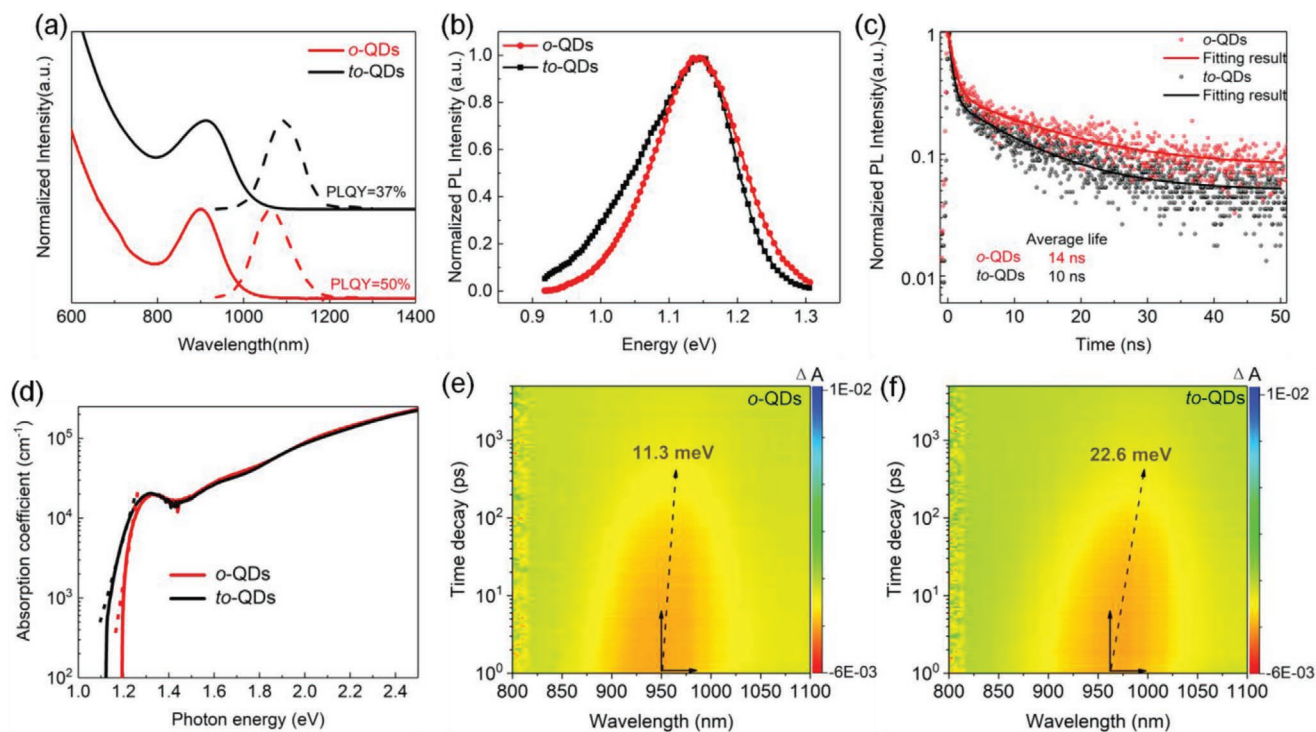


Figure 4. Optical characterizations of PbS QD solutions and solids. a) Absorption and PL spectra, and PL QY of *to*-QD (black) and *o*-QD (red) solutions before ligand exchange. b) Steady-state PL spectra, c) time-resolved PL decay, and d) Urbach energies of the two QD solids. The spectrotemporal (TA) maps for e) *o*-QD and f) *to*-QD solids.

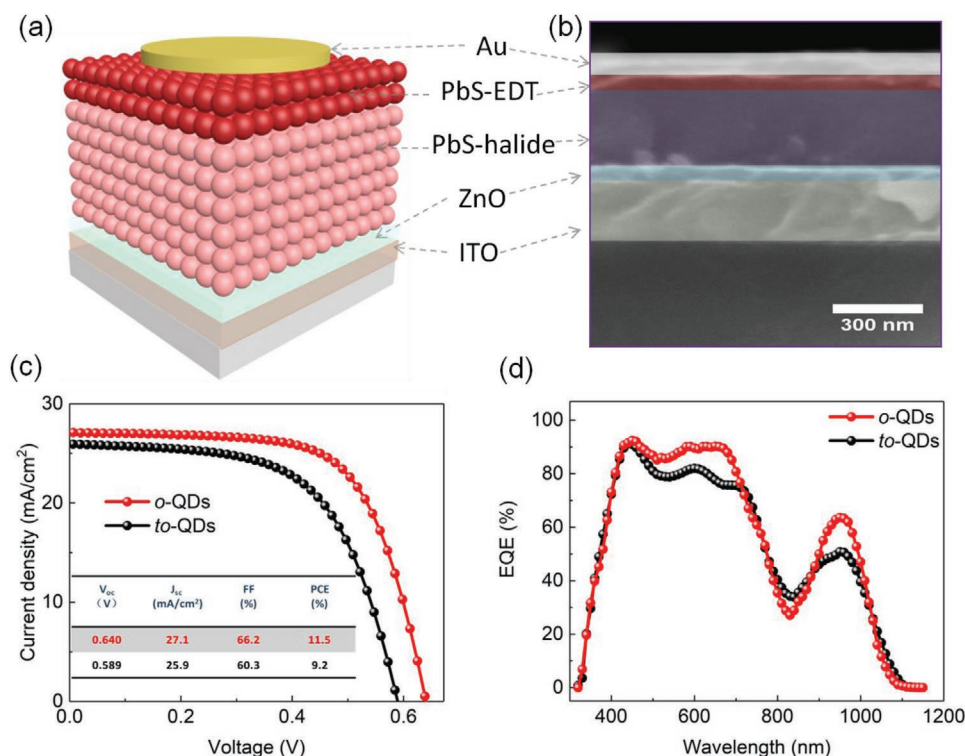


Figure 5. Device structure and performance of the PbS QD solar cells. a) Diagram of the device structure and b) cross-sectional SEM image of a typical device with characteristic layers indicated. c) J - V curves and d) EQE spectra of the champion devices based on *o*-QDs (red) and *to*-QDs (black), respectively.

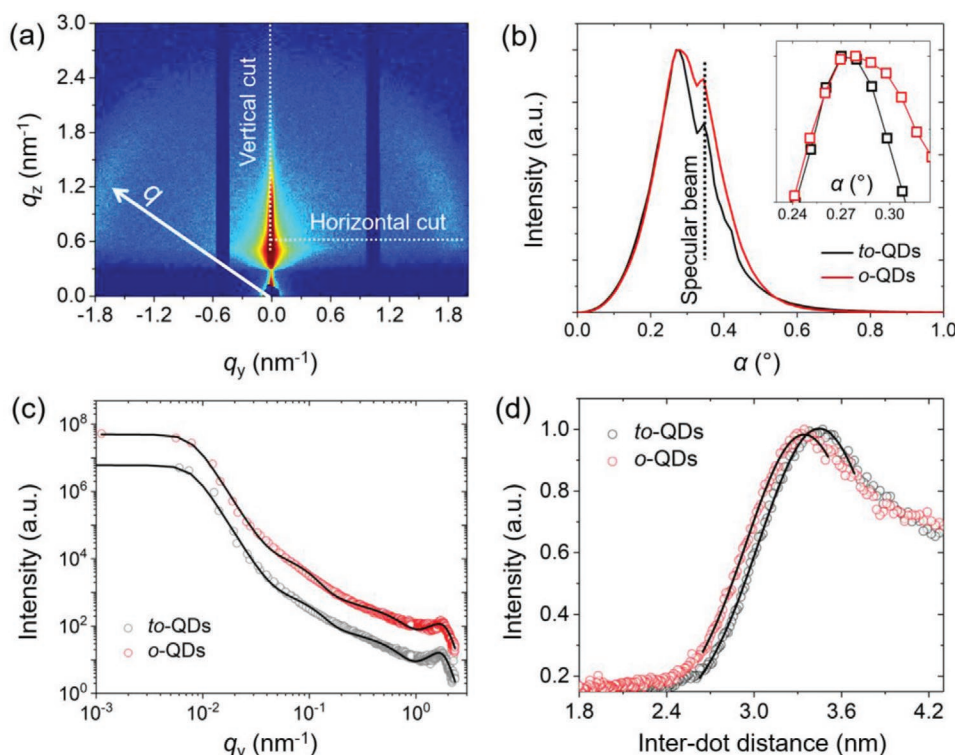


Figure 6. Particle packing behavior in the PbS QD solids. a) 2D GISAXS data of halide ligand exchanged o-QD solid with notes of q direction for integration in vertical and horizontal line cuts. b) Vertical line cuts at $q_y = 0$ nm⁻¹, c) horizontal line cuts at the Yoneda peak position, and d) interdot distance derived from the azimuthal integration for o-QD solid (red) and to-QD solid (black).

(FF) of 66.2%. The statistical results of 45 devices based on the o-PbS QDs are given in Table S2 in the Supporting Information. The peak external quantum efficiency (EQE) in the infrared region of our champion o-PbS QD solar cell reaches 64%, as shown in Figure 5d (the EQE of to-PbS QD device is also shown for comparison). The J_{SC} calculated by integrating the EQE spectrum is 26.8 mA cm⁻², which is in good agreement with the data value obtained from the $J-V$ measurement. Compared to the to-PbS QD solar cells, the predominant improvement of the o-PbS QD-based devices is V_{OC} , which is 8.7% higher than that of to-PbS QD-based devices. The V_{OC} improvement is consistent with the fewer traps and shallower bandtail in o-PbS QD solids, which is attributed to the smaller {100} facets.

In addition to an improved performance, the o-PbS QD-based devices also show excellent stability of air-storage (42 days) and illumination (4 h), as shown in Figure S6 in the Supporting Information. To avoid the contingency of the results, 45 devices were fabricated based on to-QDs and o-QDs, respectively, as shown in Figure S7 Supporting Information. The data shows that all statistical photovoltaic performance parameters (PCE, J_{SC} , V_{OC} , and FF) of o-QD-based devices are better than those of to-QD-based devices.

2.4. The Origins of the Device Improvement

In order to gain more details of the origins of the device improvement of the o-QDs, further characterizations are

performed. First, GISAXS is used to reveal the packing density of the ligand-exchanged QD solids. Figure 6a shows the 2D GISAXS data of the o-QD solid, while 2D GISAXS data of to-QDs is shown in Figure S8 in the Supporting Information. Figure 6b shows the vertical line cuts (at $q_y = 0$ nm⁻¹) of the 2D GISAXS data of both QD solids (o-QD and to-QD solids). The o-QDs produce a denser QD solid film, which is seen from the Yoneda peak position shifting from 0.27° to 0.30° (inset of Figure 6b).^[41–42] In order to get more details about the inner morphologies of the QD solids, horizontal line cuts are also analyzed (Figure 6c). Cylinders and spheres are used to model large structures (QD aggregations) and small structures (individual QDs), respectively, and to fit the scattering intensity distribution in horizontal direction. The fitted curves are presented as the solid lines in Figure 6c and the corresponding parameters are listed in Table S3 in the Supporting Information. The derived inner structure distribution of the to-QD solid and the o-QD solid are similar concerning large structures. However, the o-QD solids demonstrate a closer interdot distance of 3.2 nm than to-QDs (3.4 nm). The interdistances of the QDs in solids are also verified by the azimuthal integrations as shown in Figure 3d, with $d_{inter} = 3.34$ nm for o-QDs and $d_{inter} = 3.44$ nm for to-QDs. Thus, the tendency in horizontal direction is confirmed in this powder analysis. Due to the smaller interdot distance, the o-QDs have better electronic-coupling behavior than the to-QDs, which is beneficial for the charge transport.^[43] This accounts for the increased J_{SC} for the o-QD solar cells, as the better transport facilitates the collection of photogenerated

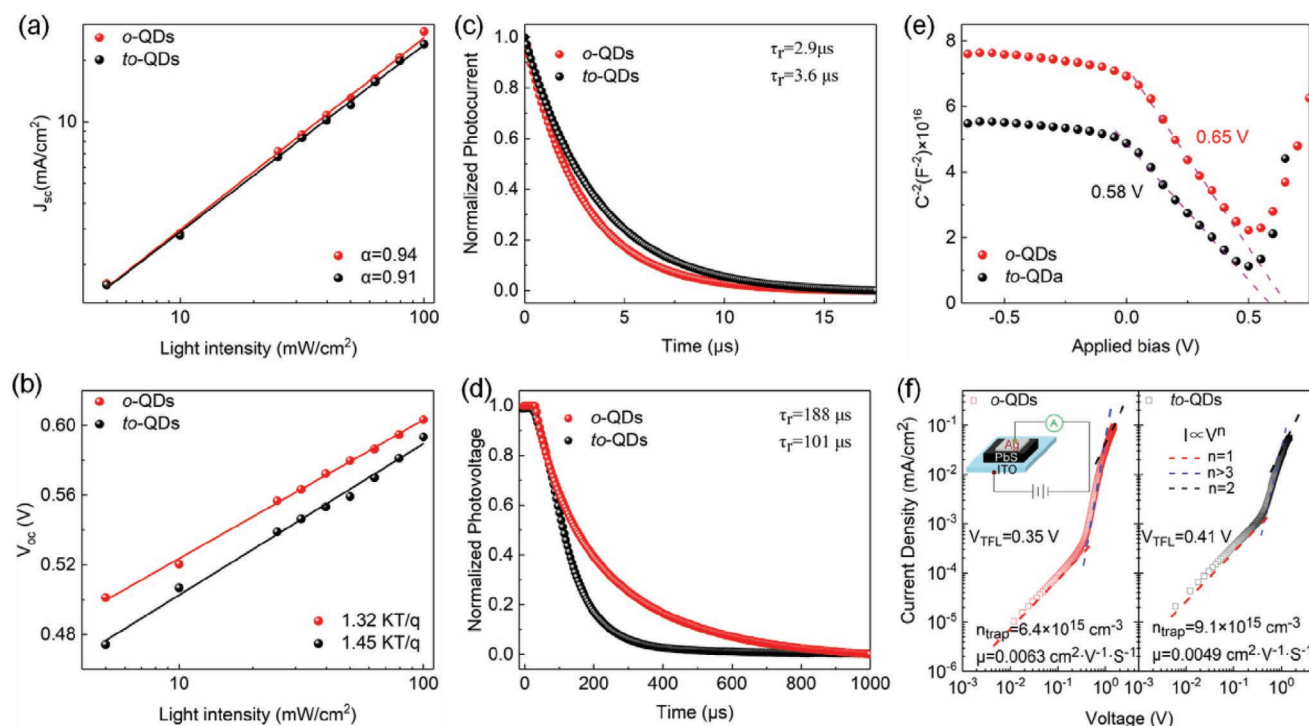


Figure 7. Device characterization of the PbS QD solar cells. a) J_{sc} and b) V_{oc} as a function of illumination intensity. c) TPC and d) TPV decay. e) Mott-Schottky analysis for calculation of the built-in potential and f) J - V curves in the SCLC analysis. Data of o-QD solid (red) are compared with data of to-QD solid (black).

charge carriers. In addition, better transport contributes to a lower series resistance, accounting for the improved FF of the o-QD solar cells.

The two types of PbS QDs have proximately the same monodispersities, demonstrated by the similar shapes of the exciton peaks as shown in Figure 4a. Size distributions obtained based on the exciton peaks following the procedure given by Tisdale and co-workers^[44] confirms the tiny difference in monodispersity, i.e., 4.9% for the o-QDs and 5.2% for the to-QDs. This tiny difference in size distribution cannot lead to the substantial enhancement of the PCE for the o-QDs. Considering the small size (≈ 3 nm), the shape difference (octahedron vs close to sphere) has negligible effect on the electronic structure and quantum confinement of the PbS QDs. Therefore, additional photophysical characterizations are carried out to gain more insights into the transport, recombination and dynamics of charge carriers. The variation of J_{sc} as a function of light intensity (I) is investigated. The dependence of J_{sc} on I can be described as $J_{sc} \propto I^\alpha$, where I is the light intensity and α is the exponential factor. The J_{sc} of devices based on both o-QDs and to-QDs show near unity intensity dependence (Figure 7a), indicating a generation-limited photocurrent.^[45–46] The dependence of V_{oc} on the light intensity is studied according to the relation: $V_{oc} \approx \frac{\eta k T}{q} \ln(I)$, where k is the Boltzmann constant, T is the temperature, q is the elementary charge, and η is the diode ideality factor. The values of η are estimated to be 1.32 and 1.45 for o-QD and to-QD solar cells, respectively (Figure 7b). The lower η indicates less trap-assisted recombination in the o-QD solids,^[47] which is consistent with

the above optical characterizations that demonstrate fewer trap states in the QD solids with smaller {100} facets. Transient photocurrent (TPC) and transient photovoltage (TPV) measurements are applied to investigate the charge carrier transport and recombination rates in devices, respectively.^[48] The TPC measurement shows a smaller lifetime for o-QD devices (Figure 7c), implying a better transport and a higher carrier collection efficiency in the o-QD devices. This result is in line with the GISAXS experiments that reveal the o-QD solids have denser particle packing behavior compared to the to-QD solids. The TPV results exhibit a longer lifetime for the o-QD devices in comparison with the to-QD devices (Figure 7d), indicating slower recombination in the o-QD solids. This finding confirms that the o-QD solids have fewer trap states in the to-QD solids. The Mott-Schottky measurements and capacitance-voltage analysis ($1/C^2$ - V) are adopted to estimate the built-in voltage (V_{bi}) in the QD solar cells. As shown in Figure 7e, the V_{bi} in o-QD- and to-QD-based devices are 0.65 and 0.58 V respectively, which highly agrees with the values of V_{oc} . Compared to the to-QD-based devices, the o-QDs solar cells show a higher V_{bi} , demonstrating a larger built-in electric field. Therefore, a higher carrier collection efficiency is expected, accounting for the higher J_{sc} of the o-QD solar cells. Finally, space-charge-limited current (SCLC) measurements^[49] confirm that the o-QD solids have larger mobility and a lower trap density than the to-QD solids, as shown in Figure 7f. As the lead halide treated PbS QD solids exhibit as n-type semiconductors,^[12,19] the SCLC measurement adopts an electron-only device structure composed of indium-doped tin oxide (ITO)/QD film/Ag.

3. Conclusion

In summary, to suppress trap states of ≈ 3 nm PbS QDs from the source, we control the balance between kinetic and thermodynamics-dominated growth conditions. The cation exchange reaction makes this control possible. By converting extremely small ZnS QDs, extremely small PbS seeds are obtained, allowing for the subsequent growth control of the small PbS QDs (≈ 3 nm). Kinetics-dominated growth is achieved at a relatively low growth temperature with a high oversaturation and the thermodynamics-dominated growth is realized at a relatively high temperature with a low oversaturation. We combine the analysis of the effect of the growth regime on the final shape for large PbS QDs, the Pb/S ratio, and the effect of QD (≈ 3 nm) shape on the packing type. The PbS QDs produced from kinetics-dominated growth have a shape of nearly octahedron due to the fact that $\chi_{100} > \chi_{111}$ for a size of ≈ 3 nm, and the PbS QDs produced from thermodynamics-dominated growth exhibit a shape of truncated octahedron due to the near isotropic growth of all the facets. Because the nearly octahedral PbS QDs have smaller {100} facets than the truncated octahedral ones, the nearly octahedral PbS QDs show fewer trap states both in isolated QDs and in electronically coupled QD solids as confirmed by the photophysical characterizations. Due to the effective suppression of trap states via facet control, the nearly octahedral PbS QDs lead to more efficient solar cells than the truncated octahedral QDs, achieving a PCE of 11.5%.

4. Experimental Section

Chemicals: Lead chloride (PbCl_2 , 99.9%), lead bromide (PbBr_2 , $\geq 98\%$), lead iodide (PbI_2 , 99%), OLA (Tech. grade, 70%), and anhydrous ammonium acetate (AA, $\geq 99.99\%$) were purchased from Sigma-Aldrich. OA (Tech. grade, 90%), 1-octadecene (ODE, Tech. grade, 90%), dimethylformamide (DMF, 99.8%), 1,2-ethanedithiol (EDT, 98%), sulfur powder (99.999%), and zinc stearate (ZnO 12.5–14%) were purchased from Alfa Aesar. Butylamine (99.5%), 1-dodecanethiol (DDT, $\geq 98\%$), and thioacetamide (TAA, $\geq 99\%$) were purchased from Aladdin. Ethanol ($\geq 99.7\%$), hexane ($\geq 97\%$), octane ($\geq 98.0\%$), acetonitrile ($\geq 99.8\%$), acetone ($\geq 99.5\%$), and thioacetamide ($\geq 99\%$) were purchased from Sinopharm Chemical Reagent. All the chemicals were used as received.

Synthesis of ZnS QDs: ZnS QDs were synthesized by a hot-injection method. Zinc stearate (4.41 g), 0.225 g (3 mmol) of TAA, 6 mL of OLA, and 50 mL of ODE were mixed in a three-neck round-bottom flask. After degassing, the mixture was heated to 140°C and maintained for 50 min. Subsequently, the mixture was cooled to room temperature, and the ZnS QDs were precipitated by centrifuge using ethanol as the antisolvent. Last, the oily ZnS QDs were obtained, then *n*-hexane and anhydrous ethanol were added again for further purification, and this process was repeated for another 1 to 2 times. Finally, the ZnS QDs were dispersed in ODE for the synthesis of PbS QDs.

Synthesis of PbS QDs: For *to*-PbS QDs, 2.919 g (10.5 mmol) of PbCl_2 and 35 mL of OLA were loaded in a three-neck round-bottom flask, and the mixture was degassed at room temperature for 30 min. Then, the temperature of the mixture was raised to 140°C under N_2 and maintained for 30 min to form the lead precursor (Pb-OLA). Subsequently, the lead precursor was set to 130°C for the injection of the ZnS QDs (diluted in 5 mL ODE). The growth was maintained for 40 s after the injection and then was quenched using a water bath. *n*-Hexane (20 mL) was injected into the reaction flask at 70°C and 28 mL of OA was injected at 40°C . At last, the reaction mixture was stirred for 10 min. The raw solution was centrifuged to remove the unreacted PbCl_2 and the supernatant

was subjected to purification using acetone as the antisolvent. For *o*-PbS QDs, the lead precursor was set to 60°C . A small amount of ZnS QDs in ODE was injected quickly to form the PbS seeds. Fifteen seconds later, a diluted ZnS QD solution (in ODE) was added dropwise using a syringe pump, until the desired exciton peak was obtained. The following procedure was exactly the same with the synthesis of *to*-PbS QDs.

Synthesis of Large PbS QDs (≈ 10 nm): For kinetics-dominated growth, the synthesis was almost the same with that of *o*-PbS QDs (≈ 3 nm), while the dropwise addition of ZnS QD solution was continued until the exciton peak reached ≈ 2100 nm. During the growth, the temperature was raised slowly to 120°C and was maintained at this temperature. For the thermodynamics-dominated growth, the procedure was similar with that of kinetics-dominated growth, except that the temperature was always maintained at 180°C and a slower addition of ZnS QD solution was adopted. It should be noted that the ≈ 10 nm PbS QDs obtained from thermodynamics-dominated growth had poor size distribution due to Ostwald ripening.

PbS QD Ink Preparation and Film Fabrication: The solution-phase ligand-exchange process was carried out in air, and followed the literature with a minor modification.^[42] Briefly, 460 mg (1 mmol) of PbI_2 , 74 mg (0.22 mmol) of PbBr_2 , and 40 mg (0.04 mol) of AA were completely dissolved in 10 mL DMF. Subsequently, 80 mg of QDs in octane were added to the 10 mL of the precursor solution. After vigorously shaking for ≈ 45 s, the PbS QDs in octane completely transferred to the polar DMF phase. Octane was added to the vial to wash the QDs in DMF. Then, the QDs were precipitated by centrifuge using ethyl acetate as the antisolvent. Finally, the PbS-halide QDs were dispersed in the mixed solution of pentylamine, hexylamine, and butylamine (3:2:10 in v/v) at a concentration of 285 mg mL^{-1} for the film fabrication. The PbS-halide QD films were deposited by spin-coating at 2500 rpm for 60 s to achieve a thickness of around 350 nm.

QD Solar Cell Fabrication: The ZnO thin film was prepared according to the method in the previous literature.^[39] The absorbing layer was fabricated by spin-coating using the PbS-halide QD ink onto the ZnO thin film. Two layers of PbS QDs treated with EDT (0.02 vol% EDT in acetonitrile solution) acted as the hole transporting layer. Finally, the Au layer with a thickness of 80 nm was deposited by thermal evaporation, which worked as the top electrode. The effective area of the prepared QD solar cell was 0.09 cm^2 .

Material Characterization: The optical absorption spectra, the PL spectra, and PL decay of PbS QDs were measured with a Shimadzu UV-3600 plus spectrophotometer, a Zolix OmniFluo fluorescence spectrometer, and an Edinburgh FLS980 fluorescence spectrometer, respectively. The TEM image of the QDs was recorded using a FEI Tecnai G2 20 UTwin transmission electron microscope. The cross-sectional SEM images were obtained using a FEI Nova NanoSEM 450 scanning electron microscope. XPS using Al $K\alpha$ excitation (Thermo Fisher, EscaLab 250Xi) was used to investigate the chemical compositions of PbS QDs. For TA measurement, the femtosecond pump-probe TA measurements were performed using a regenerative amplified Ti:sapphire laser system (800 nm, 85 fs, 7 mJ) per pulse, and 1 kHz repetition rate, Coherent, USA) as the laser source and a femto-TA spectrometer (Helios, Ultrafast systems LLC, USA). The GISAXS measurements were performed at an in-house Ganesha SAXS300L instrument. The X-ray photon energy was 8.05 keV corresponding to the wavelength of 1.54 Å. The incidence angle was set at 0.35° and the sample detector distance was 1045 mm.

Device Characterization: The *J*-*V* characteristics of QD solar cell was measured under a simulated AM 1.5 (100 mW cm^{-2}) solar spectrum from a 450 W Xenon lamp (Oriol, Model 9119, Newport). The spectra of EQE were obtained using a home-made setup containing a Keithley 2400 Source Measure unit and a Newport monochromator. *C*-*V* measurements were conducted using an Agilent 4200A at a frequency of 10 kHz and an AC signal of 50 mV, scanning from -2.5 to 0 V, with a step of 50 mV. The TPV decay and the TPC decay were measured using the home-made system and the same method as in the previous literature.^[50] The trap densities and mobility of PbS QD solids were studied using the SCLC model method with an electron-only device composed of ITO/QD solids/Ag.

DFT Calculation: The DFT calculations were performed using the projector-augmented wave (PAW) method with the Perdew–Burke–Ernzerhof (PBE) functional, as implemented in the Vienna ab initio simulation package (VASP) code. The details can be found in the Supporting Information.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Acknowledgements

Y.X. and W.C. contributed equally to this work. This work was supported by the National Natural Science Foundation of China (61974052, 61804061, 61725401, and 61861136004), the Hubei Provincial Natural Science Foundation of China (2017CFB417), and the Fundamental Research Funds for the Central Universities (2017KFYXJ039). K.W. acknowledges the National Key Research and Development Program (No. 2017YFE0120400) and the National Natural Science Foundation of China (No. 61875082). P.M.-B. acknowledges the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) under Germany's Excellence Strategy – EXC 2089/1 – 390776260 (e-conversion). S.L. acknowledges the Postdoctoral Science Foundation of China (No. 2019M662594). W.C. is grateful for financial support from the China Scholarship Council (CSC). The authors thank the Analytical and Testing Center of Huazhong University of Science and Technology for help on TEM and SEM measurements.

Conflict of Interest

The authors declare no conflict of interest.

Keywords

facet control, quantum dots, solar cells, trap-state suppression

Received: January 20, 2020

Revised: March 2, 2020

Published online: April 6, 2020

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